6. Text Generation Using OpenAI LLM API

Develop a Python program to connect to an LLM using the OpenAI API and generate text based on a user-provided prompt.

Aim

To develop a Python program that connects to an OpenAI large language model, accepts a prompt from the user, and displays the generated text.

Algorithm

1. Import the OpenAI library and the modules needed to read the API key.

2. Read the API key from the environment or request it using hidden input. Stop if the key is empty.

3. Accept a prompt from the user and check that it is not empty.

4. Create an OpenAI client using the API key.

5. Send the prompt to the model using client.responses.create().

6. Read response.output_text and display the generated text.

7. Handle authentication, quota, connection, and other API errors.

Requirements and Setup

Python 3.9 or later, an internet connection, and an OpenAI API key with access to the selected model and available API quota.

Install the Python package:
python -m pip install --upgrade openai

Save the program as text_generation.py and run:
python text_generation.py

Python Program

This program sends the user prompt to the API and prints the returned text. The API key is entered privately if OPENAI_API_KEY is not already set.

```
import os
from getpass import getpass
import openai
from openai import OpenAI

api_key = os.getenv("OPENAI_API_KEY")
if not api_key:
    api_key = getpass("Enter OpenAI API key: ").strip()
if not api_key:
    raise SystemExit("An API key is required.")

prompt = input("Enter your prompt: ").strip()
if not prompt:
    raise SystemExit("Prompt cannot be empty.")

try:
    with OpenAI(api_key=api_key) as client:
        response = client.responses.create(
            model="gpt-4.1-mini",
            input=prompt,
            max_output_tokens=300
        )
    print("\nGenerated Text:")
    print(response.output_text or "No text returned.")
except openai.AuthenticationError:
    print("Invalid API key. Check your credentials.")
except openai.RateLimitError:
    print("Check API quota or retry after some time.")
except openai.APIConnectionError:
    print("Connection failed. Check your network.")
except openai.APIStatusError as error:
    print("API request failed:", error.status_code)
```

Input

Enter OpenAI API key: [hidden input]
Enter your prompt: Explain artificial intelligence in three simple sentences.

If OPENAI_API_KEY is already set, the program skips the API key prompt.

Output

Generated Text:
Artificial intelligence allows computers to perform tasks that usually require human intelligence. It can recognise patterns, understand language, and make predictions. Examples include virtual assistants, recommendation systems, and image recognition.

The text above is an illustrative output. Actual wording may vary between requests. A live API call was not executed for this report; the Python program was checked for valid syntax.

Program Explanation

OpenAI(api_key=api_key) creates the client used for authentication and API requests. The input() function reads the prompt. The responses.create() method sends it to gpt-4.1-mini, and output_text provides the generated text. max_output_tokens=300 limits the output token budget; it is not an exact word count.

Result

A Python program was developed to generate text from a user-provided prompt through the OpenAI API. It includes input validation and basic error handling. Successful live execution requires a valid API key, model access, network connectivity, and available quota.

References

Official OpenAI documentation:
1. Python API library
https://developers.openai.com/api/reference/python
2. GPT-4.1 mini
https://developers.openai.com/api/docs/models/gpt-4.1-mini